Benchmarking Counterfactual Explanations of Model Change with Exact Ground Truth

[author names and affiliation withheld for review]

Abstract. Counterfactual methods are used to explain how a classifier changed after an update, but without ground truth one cannot tell which method sees which change. We contribute a benchmark with exact ground truth and a linear analysis of what counterfactual differencing, gradient differences, importance differences, disagreement search and differential counterfactuals (Δ-CE: nearest inputs on which exactly one model changes its decision by a margin) can see. As predicted, gradient differences are blind to an offset (confirmed on new registered pairs) and see a rotation. A pre-registered hypothesis that Δ-CE recovers the changed factor better than gradient differences was rejected. On null updates of CNNs margin witnesses abstain (0 of 2 400 queries, nulls not filtered), while differencing, IG- and SHAP-differences report a change; on a ResNet-8, whose null updates move margins four times more, they abstain only once the margin is calibrated on validation nulls, and then detect all three offsets (exploratory). Whether decisions moved can be read from a margin score; the witness adds a verified input showing where.

1 Introduction

An update that improves accuracy can still change individual decisions in unwanted ways [11]; explaining *how* they differ is a goal of explainable AI under distributional change [3, 12], distinct from explaining a shift of the data [14]. Artelt et al. [1, 2] compare a counterfactual [8] of the same input under each model, or approximate this by a difference of gradients; Muschalik et al. [4] compare feature importances; disagreement search [5, 6, 7] looks for inputs on which two models disagree. These families assume different things about *what* changed. Benchmarks with ground truth exist for explaining one model [15, 16, 17], not for explaining a change, so this goes unchecked.

We ask which kind of change each family can see. We also include *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision while f_{old} keeps it, both by a margin κ. Δ-CE anchors disagreement search [5, 6] at a query; unlike robust counterfactuals [9, 10], which seek a point valid for both models, it seeks one valid for exactly one. We use it as a probe that should stay silent when nothing changed and, when it answers, return an input that can be checked.

Contributions. (i) A benchmark with exact ground truth for explaining model updates: five rendered factors, three injected changes, null updates, and a 4×4 grid over the rotation and threshold of the decision boundary, with code and protocol. (ii) A linear analysis separating rotation from offset, with a prediction for each family. (iii) Tests on CNNs and a ResNet-8 under a protocol fixed in advance, including follow-ups registered before their data existed; all failures are reported.

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For class y the margin is $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$. A query z_0 is valid if $m_o(z_0) \geq \kappa$ and $m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|$ if $\|z^\star - z_0\| \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models decide differently, both decisively, which a difference of two counterfactuals does not.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$. Hence models whose margins differ by less than 2κ everywhere have no witness. *Proof:* subtract the two constraints of $\mathcal{F}_\kappa^N$.

Corollary 1 (constant logit offset). If f_{new} adds b to the logit of class 1, e.g. the prior correction [13], then $h \equiv \pm b$ and $\nabla h \equiv 0$. Type-N witnesses exist only for the class whose margin b lowers and only if $|b| \geq 2\kappa$, at the points with $\kappa \leq m_o \leq |b| - \kappa$. *Proof:* insert $h \equiv -|b|$ in Lemma 1; no linearity is needed.

Linear case. For linear margins on $\mathbb{R}^p$ with independent w_o, w_n the nearest witness is unique: the ordinary counterfactual of f_{new} or the minimum-norm point with both constraints active. Write $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$ with $v \perp \hat{w}_o$, and let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$. The gradient difference of [1] is $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing $u_n - u_o$ (the two minimum-norm counterfactuals), the witness $u^\star$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference lies mainly along the new dependence v; the witness is dominated by v iff $\theta > \pi/4$ or $d_o(\sin\theta + \cos\theta) < d_n$; after projecting out $\hat{w}_o$ all three coincide up to scale. *Proof* of (a): both normals equal $\hat{w}_o$; of (b): repository.

In words, gradient differences see a *rotation* and are blind to an *offset*; type-N witnesses exist only on the side whose margin the offset lowers. The analysis was formulated after the main experiment and before the grid was run.

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0,1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}[\text{aspect} > 0.5]$ with 5% noise; the renderer is exact, so the factor change of any explanation is read without a learned oracle. All methods search the same factor space through the same renderer, so no method has access the others lack. f_{old} is a small CNN (five seeds); f_{new} is f_{old} fine-tuned on 10 000 samples from one condition (Table 1). A pair enters only if its change is visible in decisions on validation data. For the grid we measure the realised rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} by a logistic fit to its decisions and

Table 1: Injected changes; $k^\star$: ground-truth factor. [a]$\alpha \in \{0, .1, .25, .5\}$, $t \in \{.5, .53, .56, .6\}$, two seeds; four offsets enter the tests. [b]Split D, registered; 8 qualified.

Change	Adaptation data	$k^\star$	Pairs
Shortcut	colour correlated with label (0.99)	colour	3
Subgroup	labels flipped for size > 0.75	size	3
Threshold	$y = \mathbb{1}[\text{aspect} > 0.6]$	aspect	3
Null	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3
Grid[a]	$y = \mathbb{1}[\text{aspect} + \alpha(\text{colour} - 0.5) > t]$	aspect / colour	8 / 24
Offsets[b]	grid, $\alpha = 0$, $t \in \{.53, .56, .6\}$, seeds 2–4	aspect	9

Table 2: Recovery@r (found-rate) with type-N Δ-CE, split A (locked), mean over three pairs (95% CIs within ± 0.035); last row: found-rate on the 8 null pairs.

	Δ-CE	CF-diff	Grad-diff	Imp-diff	Seg-search	No-prox	CF-new
Shortcut	0.38 (0.84)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
Subgroup	0.53 (0.81)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
Threshold	0.47 (0.47)	0.76 (0.90)	0.07 (1.0)	**1.00**	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null	**0.000**	0.994	1.0	1.0	0.000	0.000	0.996

analyse at these values.

Metrics and methods. Per pair: $N = 300$ valid test points. Recovery@r is the share of points with a valid explanation within the budget whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0). Recovery uses type-N witnesses only (pre-declared); a query whose class the update raises (Threshold, class 0) has none and counts 0. All counterfactual methods share one solver (Adam, then SLSQP), the points and the budget $r = 0.54$, set once on validation. We compare Δ-CE ($\kappa = 0.5$) with **CF-diff**, counterfactual differencing [1]; **Grad-diff**, the gradient difference [1]; **Imp-diff**, a difference of permutation importances on the test data [4], one vector per pair; **Seg-search**, a line search along segments from the query to training points, after [5]; **No-prox**, Δ-CE without the proximity term (the DeepXplore objective [6]); and **CF-new**, the counterfactual of f_{new} alone. DiG-IN [7] searches a learned generator for disagreement; on an exact manifold its objective reduces to No-prox. Section 4.4 adds IG- [18] and SHAP-differences [19].

Protocol. Every test split served at most one confirmatory test, after all choices were fixed: split A is the locked split of Table 2, split B the new data of the tests formed after the first results, split C the data of the type-O analysis, split D the data of three follow-ups registered [20] after the earlier results and before split D was generated. Bootstraps resample points within pairs, so their intervals do not cover model-to-model variation; Table 3 therefore also counts pairs. The rest is descriptive. A digit tier and a label-shift update failed qualification.

Table 3: Tests, grouped as in Section 4 (95% CIs; †90%). Pairs: pairs on the claimed side / pairs. expl. = exploratory. K: D = follows from a lemma, proposition or definition; R = risky prediction. When: pre = before any result; post = after the first, on split B; grid = before the grid run; new = later, new data; reg = registered [20]. Holm families: *, ‡, §. ¶Run after its registered cost rule had excluded it. Proj.: $\hat{w}_o$ projected out.

Claim	Data	Estimate	p	Pairs	Status	K, when
Abstention						
No witness on nulls	Null, B; Adult	0/2 400; 0/1 500		8/8; 5/5	confirmed	D, post; new
No witness on unfiltered nulls	Null, D	0/2 400		8/8	confirmed	R, reg
No witness on nulls	ResNet-8, D¶; E, $\kappa^\star$	180/200; 0/200		0/2; 2/2	not; expl.	R, reg; new
CF-diff (no cut) flags $\lvert b\rvert < 2\kappa$	Logit offsets	0.961 [0.955, 0.967]	$2{\cdot}10^{-4}$ §		confirmed	D, new
Offsets						
Grad-diff < 0.2	Grid, $\alpha = 0$	0.081 [0.066, 0.096]	$4{\cdot}10^{-4}$ ‡	4/4	confirmed	R, grid
Grad-diff < 0.2	Offsets, D	0.033–0.093	0.004 (sign)	8/8	confirmed	R, reg
Grad-diff offset-free	Grid	-0.098 $[-0.143, -0.053]^\dagger$	0.94 ‡		not conf.	R, grid
Δ-CE $>$ Grad-diff	Threshold, B	$+0.31$ [0.27, 0.35]	$2{\cdot}10^{-4}$ *	3/3	confirmed	D, post
Rotations						
Grad-diff rises with α	Grid	$+0.57$ [0.53, 0.61]	$4{\cdot}10^{-4}$ ‡	$\rho = 0.89$	confirmed	R, grid
Proj. Δ-CE $\approx$ Grad-diff	Shortc. B; grid	-0.30; -0.35 (CIs ± 0.03)†	1 *; 1 ‡		not conf.	R, post; grid
Normal at $z^\star$ halves gap or more	Grid, new	$-0.19 \to +0.03$ [0.01, 0.05]†	$6{\cdot}10^{-4}$ §	4/4	confirmed	R, new
Δ-*CE versus gradient differences*						
Δ-CE $>$ Grad-diff	Subgroup	-0.09 $[-0.13, -0.05]$		0/3	rejected	R, pre
Tie ($\lvert\Delta\rvert \le 0.1$)	Shortc., Thresh.	$\lvert\Delta\rvert = 0.17$–$0.53$			not conf.	R, pre

4 Results

4.1 Abstention and detection

No disagreement search returns a point on a null update (Table 2); on the 8 null pairs and on five null updates of an MLP on Adult no witness exists, while CF-diff attributes a change to 97.7–100% of points. As these nulls were admitted only without decisive disagreement on validation, close to the witness criterion, we registered 8 new null pairs admitted without any check (split D): again no witness on any of 2 400 queries, CF-diff attributes a change to 98.3–99.7%. The margin is what makes the probe silent: with $\kappa = 0$ witnesses appear on 83% of the split-A null queries. On the registered offsets the witness rate rises with the largest margin change sampled in the r-ball (from 0.02 below 0.8 to 1.00 above 1.5; descriptive): the probe reads decisive change, with κ as its unit.

Does abstention cost sensitivity? We cut each score on the nulls to flag none (or 5%) and count flagged lowered-class queries. For a constant logit offset b, witnesses appear from $|b| = 1.1$ on every lowered-class query (Corollary 1), CF-diff flags at most 0.002 up to $|b| = 2$; below 2κ the witness misses real changes by design (3–8% of decisions flip at $|b| \le 0.9$). The plain margin score $|h(z_0)|$ flags all of them from $|b| = 0.9$ at 5%. On the 8 registered fine-tuned offsets, at zero false positives, witnesses exist for 0.17–1.00 of lowered-class queries, the margin score flags 0.00–0.54 and CF-diff none (0.03–0.18 at 5%). The witness exceeded the margin score by 0.2 in 7 of 8 pairs (not in the one with the smallest realised shift), not in all as our registered criterion required, so we claim no advantage in telling *whether* decisions moved; a margin score does that cheaply. What the witness adds is *where*: a verified input near the query.

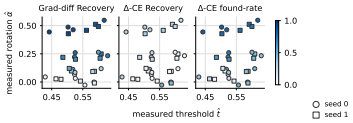


Fig. 1: Grid results at the measured rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} (grid test split), on one colour scale from 0 to 1. Hollow: no $k^\star$; marker shape: seed of f_{old}.

4.2 Offsets

Gradient differences are blind to an offset, as Proposition 2(a) predicts: on the pure offsets of the grid their Recovery is 0.081 (below 0.2 in each of the 4 pairs), and on the 8 registered offsets of split D 0.03–0.09 (8 of 8 below 0.2; sign test $p = 0.004$). On Threshold Δ-CE beats them (0.41 vs. 0.10 on split B; positive for each seed). Type-N witnesses exist only on the side the offset lowers (Threshold: class 1), hence 0.47 in Table 2; type O covers the other side (split C).

4.3 Rotations

Gradient differences see a rotation, increasingly with its angle: Grad-diff Recovery rises from 0.36 at $\alpha = 0.1$ to 0.93 at $\alpha = 0.5$ (Spearman $\rho = 0.89$ over the 24 pairs, permutation $p = 10^{-4}$; Fig. 1), while CF-new recovers colour on ≤ 0.03 of points. The projection equivalence of Proposition 2(b) holds only with $\hat{w}_o$ taken at the witness (Table 3). Where the witness condition of Proposition 2(b) holds at z_0, witnesses recover colour on 0.64 of points, elsewhere on 0.04 (descriptive).

4.4 Other baselines and a second architecture

With predictions logged beforehand (split A), IG-diff and SHAP-diff (exact Shapley values) recover colour on Shortcut (0.93) like Grad-diff, and name aspect on Threshold (0.73–0.75 vs. 0.07), against the predicted blindness. The nulls show that this is no signal: there IG- and SHAP-diff name aspect on 57% and 71% of queries (Grad-diff 5%), and, predicted for new data, IG-diff on 42% of the 2 400 unfiltered split-D null queries (Grad-diff 4%). Imp-diff on test data ranks aspect first in all 9 change pairs (Table 2), and $k^\star$ on adaptation data. With ten times the segments Seg-search finds witnesses for 0.26–0.47 of queries (predicted ≤ 0.2); on all 955 of them Δ-CE finds one too, nearer or equal on 953.

A registered ResNet-8 replication (split D, 100 queries per pair; run after its cost rule had excluded it) shows that κ is model-specific: null fine-tuning moves its margins four times more (median largest $|h|$ in the r-ball 1.93 vs. 0.51) and witnesses appear on 0.90 of null queries. Calibrating κ on validation nulls (smallest value with $\leq 5\%$ null witnesses, rule fixed in advance; 0.5 for the CNN; grid extended after $\kappa \leq 2$ failed) gives $\kappa^\star = 2.5$. On split E no null query has a witness (0/200); witnesses exist for 0.78–1.00 of lowered-class offset queries (margin score ≤ 0.05, CF-diff ≤ 0.03 at zero false positives). Recovery stays

0.36–0.40; Grad-diff is 0.32–0.36 on offsets (0.20–0.30 on nulls) and 0.06–0.10 on rotations.

5 Discussion and limitations

For *where*, we have partial checks, not a score: each witness is verified by definition, the search misses few and returns nearer witnesses than No-prox and Seg-search. **Limitations.** (1) The data are synthetic and the manifold map exact; a learned generator was not tested, and the tabular test covers abstention only. (2) κ is in logit units, exact only for a constant offset, and must be calibrated on null updates of each model family (0.5 for the CNNs; 2.5 for the ResNet-8, after an extended search). (3) Witnesses see only decision changes on the manifold within the budget, without a certificate; Recovery scores one notion of change. (4) Several tests were formed after earlier results; protocol changes were logged with dated commits before their data existed (except one analysis-code correction and the extended ResNet κ grid); log, proofs, code: [repository link].

References

[1] A. Artelt et al. Contrastive explanations for explaining model adaptations. arXiv, 2021.
[2] A. Artelt et al. Contrasting explanations for understanding and regularizing model adaptations. *Neural Process. Lett.*, 2022.
[3] B. Hammer, E. Hüllermeier. Interpretable machine learning: on the problem of explaining model change. *Workshop Comput. Intell.*, 2021.
[4] M. Muschalik et al. Agnostic explanation of model change based on feature importance. *KI* 36:211–224, 2022.
[5] X. Renard et al. Understanding prediction discrepancies in machine learning classifiers. arXiv, 2021.
[6] K. Pei et al. DeepXplore: automated whitebox testing of deep learning systems. *SOSP*, 2017.
[7] M. Augustin et al. DiG-IN: diffusion guidance for investigating networks. *CVPR*, 2024.
[8] S. Wachter et al. Counterfactual explanations without opening the black box. *Harv. J. Law Tech.*, 2017.
[9] F. Hamman et al. Robust counterfactual explanations for neural networks with probabilistic guarantees. *ICML*, 2023.
[10] M. Pawelczyk et al. On counterfactual explanations under predictive multiplicity. *UAI*, 2020.
[11] S. Yan et al. Positive-congruent training: towards regression-free model updates. *CVPR*, 2021.
[12] M. Kaden et al. Reliable counterfactuals for machine learning models – current aspects and perspectives. *ESANN*, 2026.
[13] M. Saerens et al. Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure. *Neural Comput.*, 2002.
[14] S. Kulinski, D. I. Inouye. Towards explaining distribution shifts. *ICML*, 2023.
[15] C. Agarwal et al. OpenXAI: towards a transparent evaluation of model explanations. *NeurIPS*, 2022.
[16] R. Hesse et al. FunnyBirds: a synthetic vision dataset for a part-based analysis of explainable AI methods. *ICCV*, 2023.
[17] L. Arras et al. CLEVR-XAI: a benchmark dataset for the ground truth evaluation of neural network explanations. *Inf. Fusion*, 2022.
[18] M. Sundararajan et al. Axiomatic attribution for deep networks. *ICML*, 2017.
[19] S. Lundberg, S. Lee. A unified approach to interpreting model predictions. *NeurIPS*, 2017.
[20] Registration of the follow-up experiments. OSF, 2026. https://osf.io/vfjtk